Random Effects Models

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1 Overview

Objectives and overview

- This material is based Chapter 11–12 on the textbook "A First Course in Design and Analysis of Experiments" (Oehlert, 2000).
- Chapter outcomes include:
  - Students will be able to distinguish between a random effect and a fixed effect.
  - Students will be able to identify when random effects are appropriate.
  - Students will know how to implement random effects and mixed effects models.
  - Students will gain some familiarity with the `lme4` package in R.

Introduction

- So far, the treatment effects that we have been working with can be classified as *fixed-effects*: we wish to estimate the effect of specific (fixed) treatments.
- What if the treatments themselves are a random selection from a larger population?
- **Example:** A farmer has 100 pieces of equipment, and would like to know how the equipment used may impact crop-yield. Rather than testing all 100 pieces of equipment, they *randomly select* g pieces of equipment, and performs a CRD with these equipment.

Random Effects

A random effect is appropriate to model treatments that are a *random sample* from a population of potential treatments.
Random effects are useful if the focus of inference is on the *population*, not the individual treatment effects.

2 Models for Random Effects

Example: Carton factory

- We will build intuition of Random Effects models and designs using an example.

Carboard company
Suppose a company makes cardboard cartons for storing goods. They have 50 machines to make cartons, and want to understand the variation in quality for finished products.
They choose 10 machines at random from the 50, and make 40 cartons on each machine. The resulting cartons are tested for strength.

- This is a completely randomized design (CRD), with $g = 10$ treatments (the different machines), and $n_i = 40$, for a total of $N = 400$ EUs.
- The model we have been using for this type of data is:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \quad \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2).$$

- We are still going to use a very similar model. However, we are not as interested in the exact α_i for the 10 randomly chosen machines. Instead, we want to make inference on the larger population.
- We could proceed as usual, but that only tells us about the 10 specific machines that were randomly chosen, and we would get an entirely different set of machines if we repeated the experiment.
- Thus, we want to introduce a *random effect*: let the effect α_i represent a random draw from a population of possible effects:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \qquad \alpha_i \overset{iid}{\sim} N(0, \sigma^2_\alpha), \quad \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2).$$

- We no longer assume $\sum_i \alpha_i = 0$, rather $E[\alpha_i] = 0$.
- We have introduced a new parameter, σ^2_α, which describes the spread of the effects.
- *We assume that the two random components are independent.* That is, α_i and ϵ_{ij} are independent.
- If they are independent, the variance of the observed data is increased:

$$\text{Var}(y_{ij}) = \text{Var}(\mu + \alpha_i + \epsilon_{ij}) = \sigma^2_\alpha + \sigma^2.$$

- The parameters σ^2_α and σ^2 are called *variance components*.
- If observations belong to the same treatment group, then they now have non-zero correlation:

$$\text{Cor}(y_{ij}, y_{ik}) = \begin{cases} 1 & j = k \\ \sigma^2_\alpha/(\sigma^2_\alpha + \sigma^2) & j \neq k \end{cases}$$

- We call this within treatment group correlation the *intraclass* correlation.

Key summary

- In a random effects model, the parameters are μ, σ^2, and σ^2_α.
- We are not interested in making inferences about the specific α_i and ϵ_{ij}, which are treated as random variables, not parameters.
- Inference wise, we are now interested in confidence intervals and hypothesis tests regarding μ and σ^2.

Hypothesis Testing

- If we are testing for population effects based on our random sample, there are multiple ways to do this formally.
- The hypothesis we are generally interested in testing in this scenario is:

$$H_0 : \sigma_\alpha^2 = 0, \quad \text{vs} \quad H_0 : \sigma^2 > 0.$$

- Surprisingly, this leads to the same F-tests that we have been using before!
- Specifically, we have said that $E[\alpha_i] = 0$; thus, if H_0 is true, then $\text{Var}(\alpha_i) = 0$, and by necessity, $\alpha_i = 0$. Thus,

$$\sigma_\alpha^2 = 0 \iff \alpha_i = 0 \iff \mu_i = \mu_j \, \forall (i, j).$$

Example data in R

- The following example uses data synthetically generated with $\alpha_i \sim N(0, 2^2)$, $\epsilon_{ij} \sim N(0, 0.5^2)$, and $\mu = 10$.

```
synthData <- read.csv("synthData.csv")
mod <- aov(response ~ factor(trt), data = synthData)
summary(mod)

            Df Sum Sq Mean Sq F value   Pr(>F)
factor(trt)  3  94.21  31.404   40.66 2.46e-10 ***
Residuals   28  21.63   0.772
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Above, we see a significant (random) treatment effect, rejecting the hypothesis that $\sigma_\alpha^2 = 0$.

- Note in this problem, the F statistic that is used matches the R output:

$$F_{DF_A, DF_E} = \frac{MS_A}{MS_E}.$$

  This will *not* be the case for factorial models or if the data are unbalanced.

Estimation variance components

- In perfectly balanced data, we *can* get point estimates for σ_α^2; this follows a similar approach to estimating σ^2, like we did back in module 3.
- Recall: $\hat{\sigma}^2 = \text{MS}_\text{E}$.
- If the data are balanced, *then*:

$$\hat{\sigma}_\alpha^2 = \frac{MS_\text{A} - MS_\text{E}}{n}$$

- In the previous R example, this is $(31.404 - 0.772)/8 = 3.8290$
- **NOTE:** This can be problematic if $MS_A < MS_E$, because we will get negative estimates! The approach we will discuss later does not have this same problem.

3 Estimation: REML and LRT

Beyond ANOVA: Likelihood Based Inference

- Classical random effects estimation relies on Expected Mean Squares (EMS) from ANOVA tables (See Oehlert (2000, Sections 11.3–11.6)).
- These approaches can be computed by hand, and would naturally extend our workflow with the `aov()` function and the `emmeans` package.
- Unfortunately, these approaches have their weaknesses for random (and mixed) effects models.
- Our models will remain the same, but how we analyze them will shift, keeping in-line with modern practices.
- For the type of models considered so far (fixed-effects), the two approaches are the same.
- Modern software relies on *Likelihood* methods.
- **Maximum Likelihood (ML)** finds the parameter estimates (mean and variances) that make the observed data most probable; confidence intervals are obtained using likelihood ratio tests.
- However, standard ML has a flaw when estimating variance components: it does not account for the degrees of freedom used to estimate the fixed effects (like the grand mean).
- This makes standard ML variance estimates *biased* (they underestimate the true variance).

  Example: For a normal distribution model, $y_{ij} \sim N(\mu_i, \sigma^2)$, the MLE of (μ, σ^2) is:

$$\left(\hat{\mu}_{MLE}, \hat{\sigma}^2_{MLE}\right) = \left(\bar{Y}_i, \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2\right).$$

  However, you may recall from you intro stats class that we usually don't estimate variance that way, instead we divide by n_1:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

  That's because the MLE for the variance component is biased low: on average, our estimate will be slightly smaller than the truth.

Restricted Maximum Likelihood (REML)

- To fix the bias in standard ML, modern software uses *Restricted Maximum Likelihood (REML).*
- REML mathematically removes the fixed effects from the likelihood equation before estimating the variance components.
- By accounting for the degrees of freedom lost in estimating fixed effects, REML provides *unbiased* estimates of σ^2 and σ^2_α.

  Example: building on the normally distributed model from before, the MLE for σ^2 was biased low:

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

  The REML estimate for this model is unbiased:

$$\hat{\sigma}^2_{REML} = s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- Additionally, REML handles unbalanced data effortlessly and prevents the impossible "negative variance" estimates that can sometimes occur when calculating variance using classical EMS methods.

Estimating variance components in R

- If we are primarily interested in the variance components σ^2_α, we will want to get the REML estimate for this component, as it is unbiased.
- We will shift from our `aov()` workflow and use the `lmer` function from the `lme4` package to do this in R:

```
# install.packages("lme4")
library(lme4)
lmer(response ~ 1 + (1 | randomEffect), data = data)
```

On the next slide, we will use this for the same `synthData` dataset we used the `aov` function for.

Example: Synthetic data

```
# Old Code:
# aov(response ~ factor(trt), data = synthData)
synthData$trt <- factor(synthData$trt)
lmer(response ~ (1 | trt), data = synthData)

Linear mixed model fit by REML ['lmerMod']
Formula: response ~ (1 | trt)
   Data: synthData
REML criterion at convergence: 94.5478
Random effects:
 Groups   Name        Std.Dev.
 trt      (Intercept) 1.9568
 Residual             0.8788
Number of obs: 32, groups:  trt, 4
Fixed Effects:
(Intercept)
      12.78
```

The important thing to note in the output above is that the estimated values match what we did by hand with the `aov()` output. Specifically, our estimate for treatment (`trt`) had $\hat{\sigma}_\alpha = 1.9568$; if we square it, we get the same value as the by-hand calculation of $\hat{\sigma}^2_\alpha = 3.829$.

Example: Penicillin

- We will first consider the built-in `Penicillin` data in R, which contains the reduced diameter of bacterial growth, after penicillin application. Each measurement is taken on a specific `plate`, which is our random effect.

```
data("Penicillin")

output <- lmer(
  diameter ~ 1 + (1 | plate),
  data = Penicillin, REML = TRUE  # default
)
```

- What is the notation here doing? `1 + (1 | plate)`.
  - The model we are fitting is $y_{ij} = \mu + \alpha_i + \epsilon_{ij}$.
  - The `1` in the beginning is not strictly needed, but represents the grand mean μ.
  - The `(1 | plate)` represents the random effect. In `lme4`, the syntax `(A | B)` implies that B is the random effect grouping variable, and A is the effect that is random. For a simple random effects model, the mean (`1`) is always the thing that is random.
- Now let's look at the output.

```
summary(output)

Linear mixed model fit by REML ['lmerMod']
Formula: diameter ~ 1 + (1 | plate)
   Data: Penicillin

REML criterion at convergence: 613.3

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.41583 -0.55523  0.01212  0.59232  1.95241

Random effects:
 Groups   Name        Variance Std.Dev.
 plate    (Intercept) 0.09509  0.3084
 Residual             4.03333  2.0083
Number of obs: 144, groups:  plate, 24

Fixed effects:
            Estimate Std. Error t value
(Intercept)  22.9722     0.1788   128.5
```

Comparing to `aov`

- Since the data are balanced, the estimated parameters of this new `lmer` function will match what we can compute with `aov`:

Here, we can tell $n = 6$ because there are $g = 24$ plates, so $120 + 24 = 144$ total observations. Since it is balanced, there are $n = 144/24 = 6$ observations per group.

```
# Compute (MS_plate - MS_E) / n, n = 6
summary(aov(diameter ~ plate, data = Penicillin))

             Df Sum Sq Mean Sq F value Pr(>F)
plate        23  105.9   4.604   1.141  0.313
Residuals   120  484.0   4.033
```

Hypothesis testing

- The `lmer` function uses REML to get parameter estimates, i.e., $\hat{\sigma}^2_\alpha$.
- If we want to perform a hypothesis test for $\sigma^2_\alpha > 0$, we have two main options:
  1. Refit the model with `aov()`, and use the built-in F-test.

  - **Advantage:** Exact F-tests. These are more *powerful*.
  - **Disadvantage:** Only works if data are balanced, and the output P-value only matches if there is a single random factor in the model (otherwise, can compute by hand).
2. Use likelihood based inference procedures (shown below).
  - **Advantage:** Works whether or not data is balanced, and output P-values always "correct". More robust to violations of assumptions.
  - **Disadvantage:** Based on large-sample sizes, so less powerful if samples are small.

- We have already seen analysis of 1-factor random effects with `aov`, now let's do the second approach.

Confidence intervals

- We can get confidence intervals for our estimates using the `confint` function.
- The two primary methods will be `profile` (default) and `boot`.
- This computes *profile likelihood* confidence intervals, which are effectively the same as a **likelihood ratio test** (LRT).
- Mathematically, we can obtain a confidence interval using profiled restricted likelihoods, but unfortunately there's not an immediate way to do this in R; we can use `method = "boot"` for a bootstrap interval for REML estimates, or `method = "profile"` for a profile likelihood. Both should give very similar output.

```
set.seed(123)
confint(output, signame = FALSE, level = 0.95, method = "boot")
```

```
                          2.5 %     97.5 %
sd_(Intercept)|plate  0.000000  0.8138926
sigma                 1.771368  2.2218709
(Intercept)          22.656619 23.3378052
```

Hypothesis testing

- For random effects, the hypothesis that we would like to test is:

$$H_0 : \sigma_\alpha^2 = 0, \quad H_1 : \sigma_\alpha^2 > 0$$

- This is always one-sided, because σ_α^2 can't be negative.
- *Notice:* Our assumption was $\alpha_i \sim N(0, \sigma_\alpha^2)$. If H_0 is true, then $\sigma_\alpha^2 = 0$, and $\alpha_i = 0$ for all i.
- Thus, for a simple random effects model, we have the same null hypothesis and tests!

$$H_0 : \sigma_\alpha^2 = 0 \iff \alpha_i = 0 \text{ for all } i$$

- Because of this, we *could* just use a standard F-test for random effects.
- *However:* this approach does not generalize well to more complex models with random effects.
- Rather than using an F-test, we will introduce the idea of a Likelihood Ratio Test.

Likelihood Ratio Tests (LRT)

- Instead of using exact F-tests derived from ratios of Mean Squares, modern mixed modeling often uses the *Likelihood Ratio Test (LRT)* for hypothesis testing.
- The LRT explicitly compares two competing models:
  1. A **Full Model** that includes the random effect.
  2. A **Reduced Model** (Null Model) that removes the random effect.
- The test measures how much the likelihood (fit) of the data drops when the random effect is removed.

$$\Lambda = \frac{\text{Likelihood}_{\text{reduced}}}{\text{Likelihood}_{\text{Full}}}, \quad \Longrightarrow \quad -2\log\Lambda \sim \chi^2_d$$

- Using properties of logarithms, we often consider differences in log-likelihood:

$$\ell_{\text{full}} - \ell_{\text{reduced}} \sim \frac{1}{2}\chi^2_d,$$

  where d is the difference in number parameters between the models.
- The test can be extended for restricted likelihood ratio tests (RLRT); the idea is similar: fit two models (full vs reduced) using REML, compare differences in the restricted log-likelihoods.
- Rather than fitting two models manually and comparing, we will use the `lmerTest` package to do this automatically for us.

```
# install.packages("lmerTest")
lmerTest::ranova(output)  # Will do RLRT or LRT, depending on object.

ANOVA-like table for random-effects: Single term deletions

Model:
diameter ~ 1 + (1 | plate)
            npar  logLik    AIC      LRT Df Pr(>Chisq)
<none>         3 -305.81 617.63
(1 | plate)    2 -305.86 615.71 0.082153  1     0.7744
```

4 Factorial Random Effects Models

Factorial models

- Just like with fixed effects, we may want to consider random effects with multiple random treatment factors.
- Again, we use random effects when we aren't interested in the specific treatment effects chosen, but the larger population of treatment effects.

Example: Carton Factory (continued)

- Let's expand our motivating example of the Carton Factory.

Two-way random effects model
Suppose the carton factory has multiple machines and would like to know the variance in the effect of these machines, but there are also different machine operators, and their effect may be random.

- We randomly select machines and operators, and have each operator work on each each machine and produce $n = 4$ cartons each.
- We have a two-way, completely randomized factorial design.
- In the design proposed above, a two-way random effects model is natural:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk},$$

with

$$\alpha_i \overset{iid}{\sim} N(0, \sigma^2_\alpha) \quad \beta_j \overset{iid}{\sim} N(0, \sigma^2_\beta) \quad (\alpha\beta)_{ij} \overset{iid}{\sim} N(0, \sigma^2_{\alpha\beta}) \quad \epsilon_{ijk} \overset{iid}{\sim} N(0, \sigma^2)$$

- The variance of each observation is

$$\text{Var}(y_{ijk}) = \sigma^2_\alpha + \sigma^2_\beta + \sigma^2_{\alpha\beta} + \sigma^2.$$

- This idea once again can generalize to higher-order factorial models

Parameter estimation

- The parameter estimation will still happen in a similar way: we can use ML or REML.
- If we are primarily interested in the error components, REML is unbiased, and generally preferred.

```
carton <- read.csv('carton.csv')
output <- lmer(
  thickness ~ 1 + (1 | Machine) + (1 | Operator) + (1 | Machine:Operator),
  # OR: 1 + (1 | Machine * Operator)
  data = carton,
  REML = TRUE
)
```

```
summary(output)

Linear mixed model fit by REML ['lmerMod']
Formula:
thickness ~ 1 + (1 | Machine) + (1 | Operator) + (1 | Machine:Operator)
   Data: carton

REML criterion at convergence: 317.2

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.58412 -0.78134 -0.07557  0.72636  2.15223
```

```
Random effects:
 Groups           Name        Variance Std.Dev.
 Machine:Operator (Intercept)  3.782   1.945
 Machine          (Intercept)  2.083   1.443
 Operator         (Intercept) 11.024   3.320
 Residual                      1.465   1.210
Number of obs: 80, groups:
Machine:Operator, 20; Machine, 5; Operator, 4

Fixed effects:
            Estimate Std. Error t value
(Intercept)   46.671      1.838   25.39
```

Hypothesis Testing (F-test)

- Recall that the F-test is exact, and valid for testing the hypothesis $\sigma^2_A, \sigma^2_B, \sigma^2_{AB} > 0$.
- To do this, we would re-fit the model with `aov()`.
- *HOWEVER*: we must be extremely careful. The output P-values are INCORRECT, because of the factorial structure.
- Exact P-values can be computed by hand (visit office hours if you're interested). However, we will instead use the likelihood based inference methods.

  For the notes only, here are a few details regarding the "by-hand" P-value calculations.
- In the output ANOVA table, the data are balanced, meaning that the Mean Sq values that are output are correct. However, if we have factorial random effects structure, then the correct F-test does NOT match the output of `aov`. In the output ANOVA, the F-value for fixed effects is computed as MS_{effect}/MS_E. For random effects, the denominator must change. For the Factor A test, the F value is MS_A/MS_{AB}; this also impacts the degrees of freedom. Our textbook (**?**) goes into a lot of these details if you're interested. However, pure factorial random effects models are not that common in practice. We'll later go into mixed-effects models, at which point we will strictly need to use `lme4::lmer`. Because of this, it makes sense to just highlight this now, rather than going over tedious by-hand computations that are not often seen in practice.

Confidence intervals

- We can obtain confidence intervals the same way as before. This can be very slow.

```
set.seed(444)
confint(output, method = 'boot', signame = FALSE)

                                    2.5 %    97.5 %
sd_(Intercept)|Machine:Operator  1.0920933  2.746098
sd_(Intercept)|Machine           0.0000000  3.031105
sd_(Intercept)|Operator          0.1061933  5.646589
sigma                            0.9914768  1.422556
(Intercept)                     43.4999935 50.636162
```

Hypothesis testing (likelihood based)

- We can also do hypothesis testing, but it can get tricky due to the interaction effect.
- The `lmerTest::ranova` function just drops the term from the model, and compares REML (or ML). This may not be meaningful for all variables in the presence of an interaction effect.
- Instead, we often will rely on the confidence intervals.

```
# May not be meaningful for main (random) effects if interaction is present.
lmerTest::ranova(output)

ANOVA-like table for random-effects: Single term deletions

Model:
thickness ~ 1 + (1 | Machine) + (1 | Operator) + (1 | Machine:Operator)
                       npar  logLik    AIC    LRT Df
<none>                    5 -158.62 327.24
(1 | Machine)             4 -159.67 327.35  2.105  1
(1 | Operator)            4 -164.39 336.77 11.531  1
(1 | Machine:Operator)    4 -180.10 368.19 42.952  1
                       Pr(>Chisq)
<none>
(1 | Machine)           0.1468521
(1 | Operator)          0.0006846 ***
(1 | Machine:Operator)   5.61e-11 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Summary

- We fit a random effects model if we are interested in the variance components specifically, rather than specific treatment effects.

5 Mixed Effects Models

Mixed Effects: an introduction

- One of the most common ways we see a random effect is in the context of a *mixed effect model*.
- Idea: we are interested in estimated the fixed effects of specific treatments, but have factors in the study that are randomly drawn from a population.
- Mixed effect models are extremely common for *block-designs*: we have a treatment factor A we are interested (fixed effect), and a block factor B from a block-design; if we are not interested in the effect of our specificly chosen blocks, rather we want to account for the potential variation of all possible blocks that could have been chosen, then a mixed-effects models is the correct choice.

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